Chandubhai S Patel Institute of Technology

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGY AND ENGINEERING
B. TECH (IT/CE/CSE/AIML)

DEPSTAR

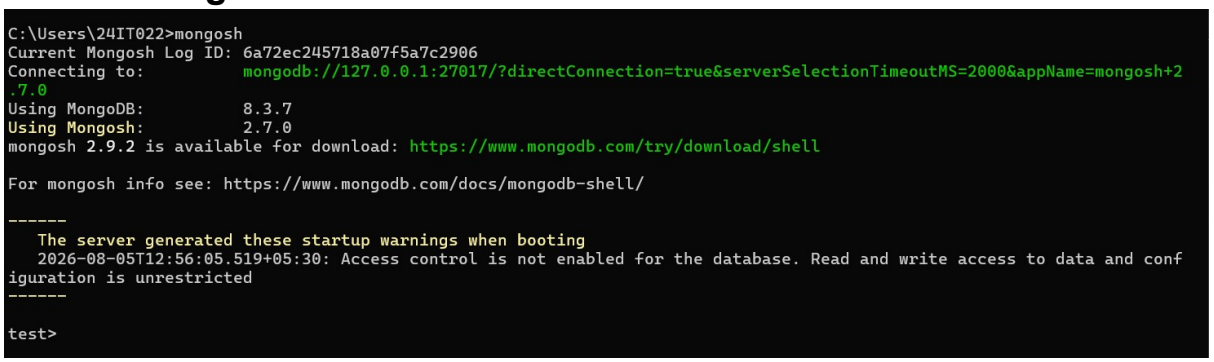
Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement:

- Extend the Task Management backend from Practical 4 by connecting MongoDB using Mongoose.
- Define a Task schema with at least 4 fields: title (String, required), description (String), completed (Boolean, default false), createdAt (Date, default Date.now).
- Replace the in-memory array from Practical 4 with real Mongoos model operations.
- Test all CRUD operations against the live database using Postman.
- Ensure validation errors are returned as structured JSON, not raw Mongoose error objects.

Screenshots:

Step 1: Start MongoDB



Step 2: Install Packages

```
PS E:\CHARUSAT\SEM 5\AWDF\Pra_4\task-manager-api>
 npm install mongoose dotenv

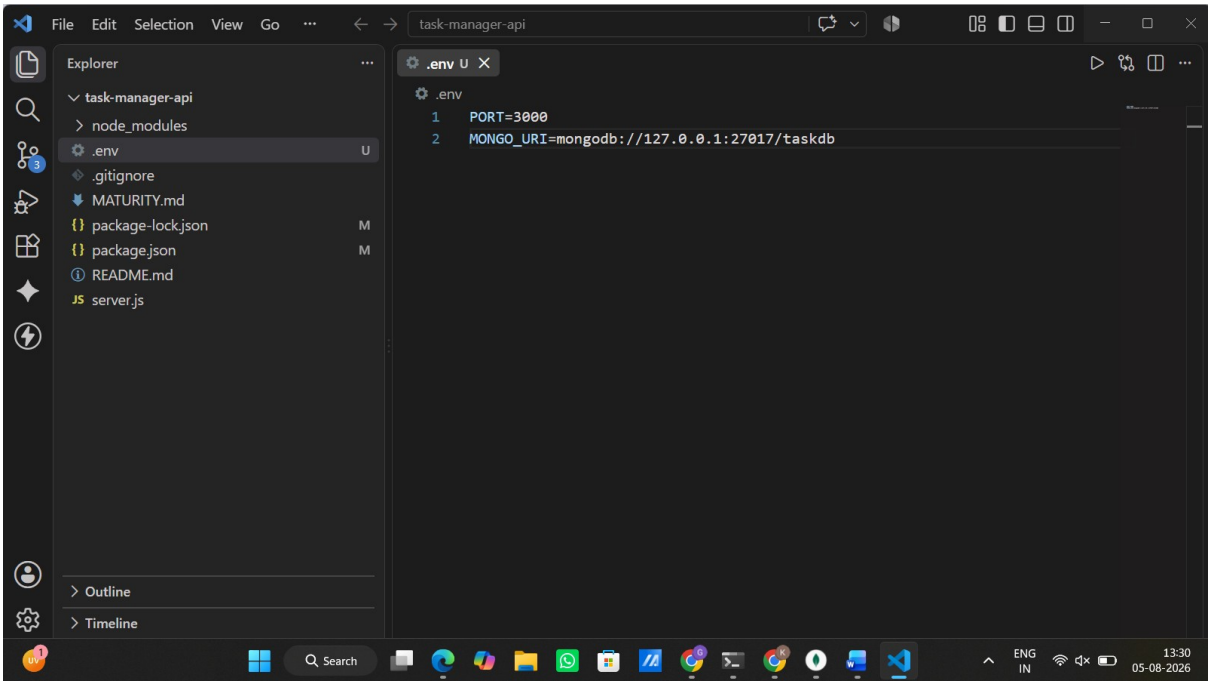
added 19 packages, and audited 87 packages in 9s

28 packages are looking for funding
  run `npm fund` for details

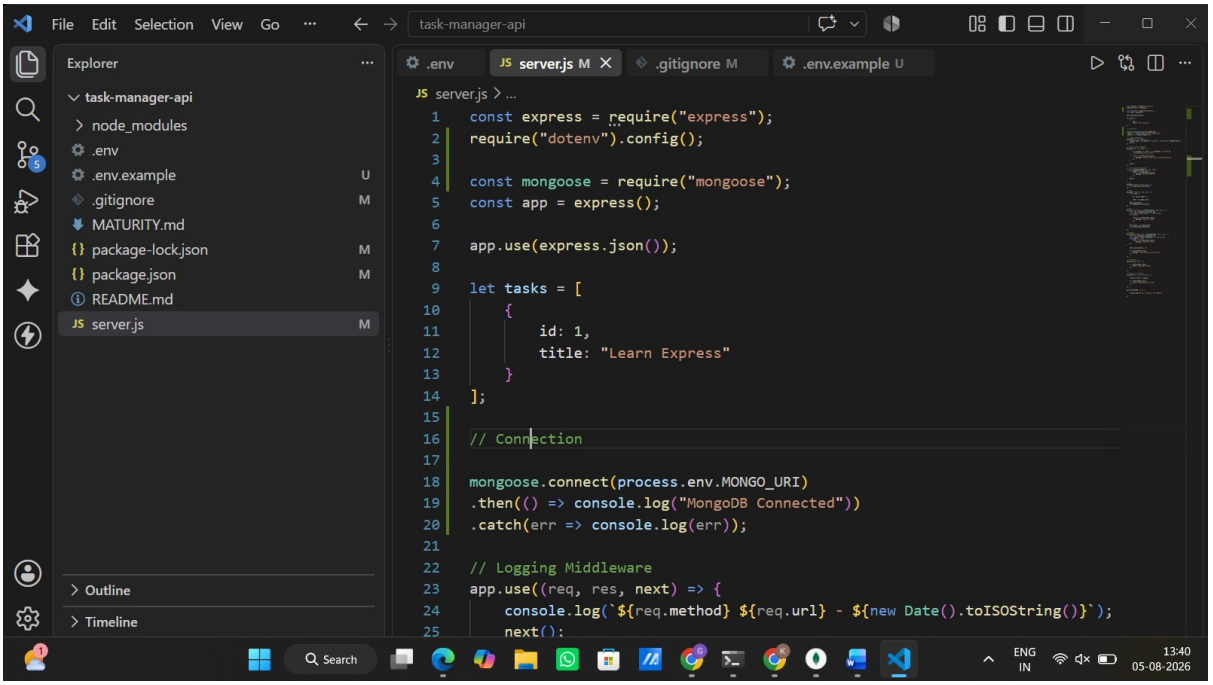
found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 11
.16.0 -> 12.0.2
npm notice Changelog: https://github.com/npm/cli/
releases/tag/v12.0.2
npm notice To update run: npm install -g npm@12.0
.2
npm notice
PS E:\CHARUSAT\SEM 5\AWDF\Pra_4\task-manager-api>
 npm list mongoose
task-manager-api@1.0.0 E:\CHARUSAT\SEM 5\AWDF\Pra
_4\task-manager-api
└── mongoose@9.9.1

PS E:\CHARUSAT\SEM 5\AWDF\Pra_4\task-manager-api>
```

Step 3: Create .env

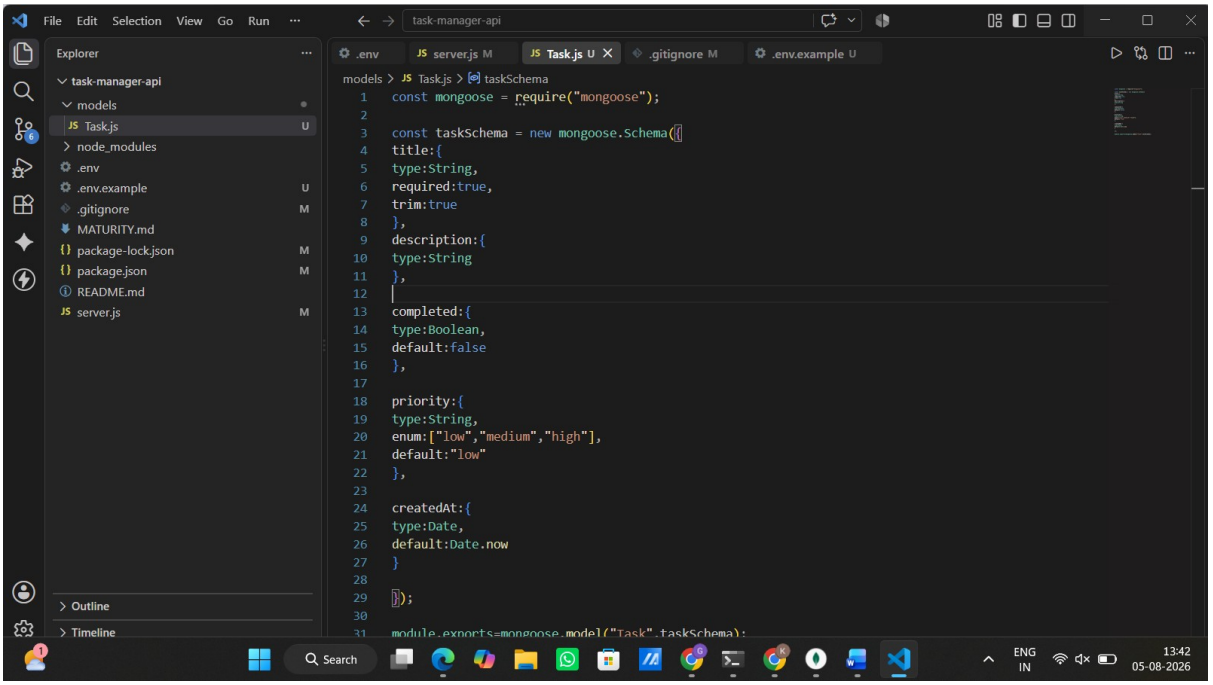


Step 4: Connect MongoDB

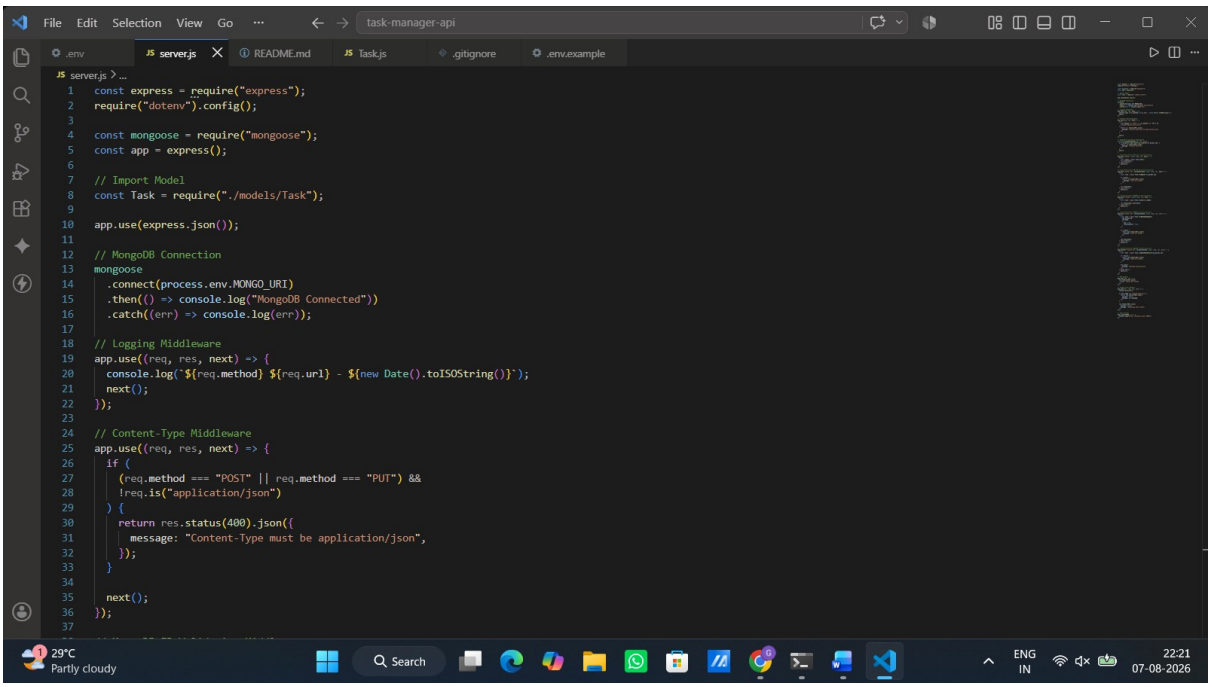




Step 5: Create Schema



Step: 6 Implement MongoDB CRUD Operations with Mongoose



```
const express = require("express");
require("dotenv").config();

const mongoose = require("mongoose");
const app = express();

// Import Model
const Task = require("./models/Task");

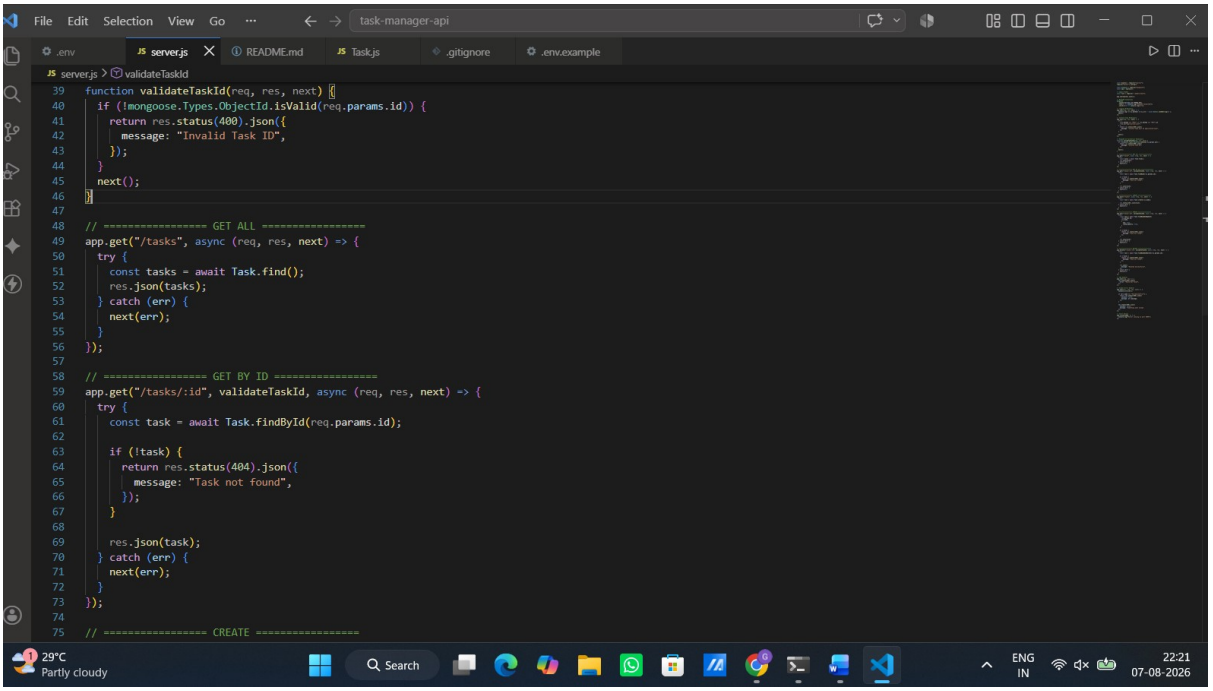
app.use(express.json());

// MongoDB Connection
mongoose
  .connect(process.env.MONGO_URI)
  .then(() => console.log("MongoDB Connected"))
  .catch((err) => console.log(err));

// Logging Middleware
app.use((req, res, next) => {
  console.log(`${req.method} ${req.url} - ${new Date().toISOString()}`);
  next();
});

// Content-Type Middleware
app.use((req, res, next) => {
  if (
    (req.method === "POST" || req.method === "PUT") &&
    !req.is("application/json")
  ) {
    return res.status(400).json({
      message: "Content-Type must be application/json",
    });
  }

  next();
});
```



```
function validateTaskId(req, res, next) {
  if (!mongoose.Types.ObjectId.isValid(req.params.id)) {
    return res.status(400).json({
      message: "Invalid Task ID",
    });
  }
  next();
}

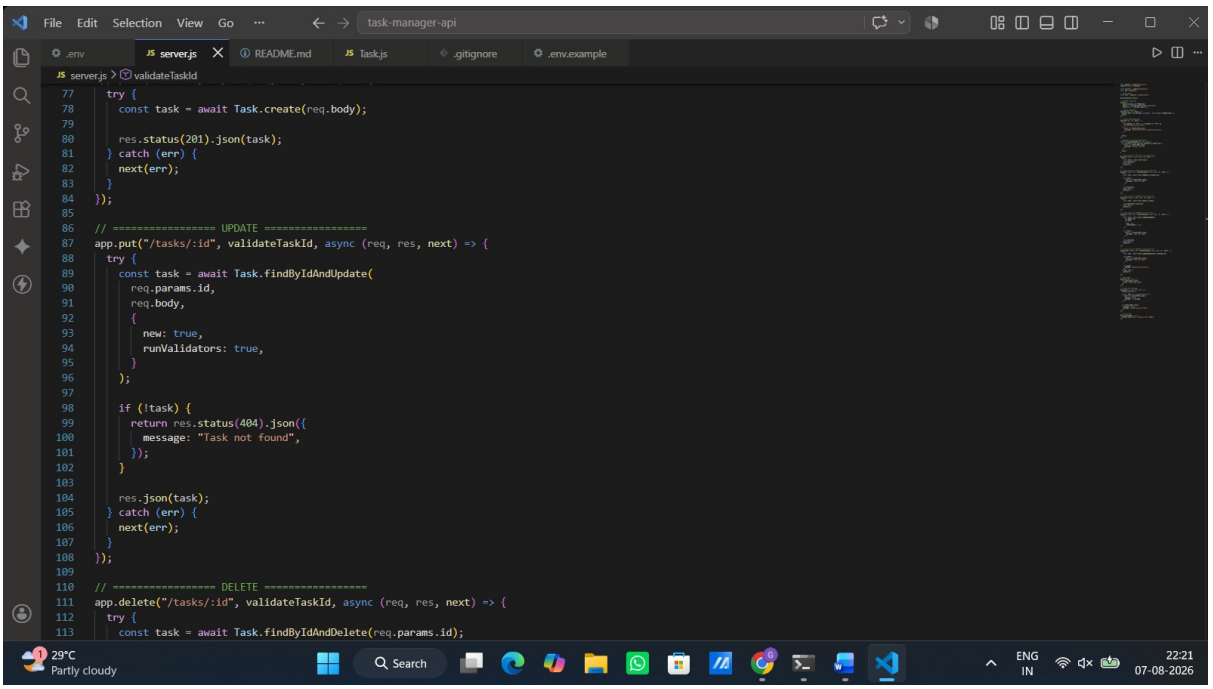
// ================ GET ALL ================
app.get("/tasks", async (req, res, next) => {
  try {
    const tasks = await Task.find();
    res.json(tasks);
  } catch (err) {
    next(err);
  }
});

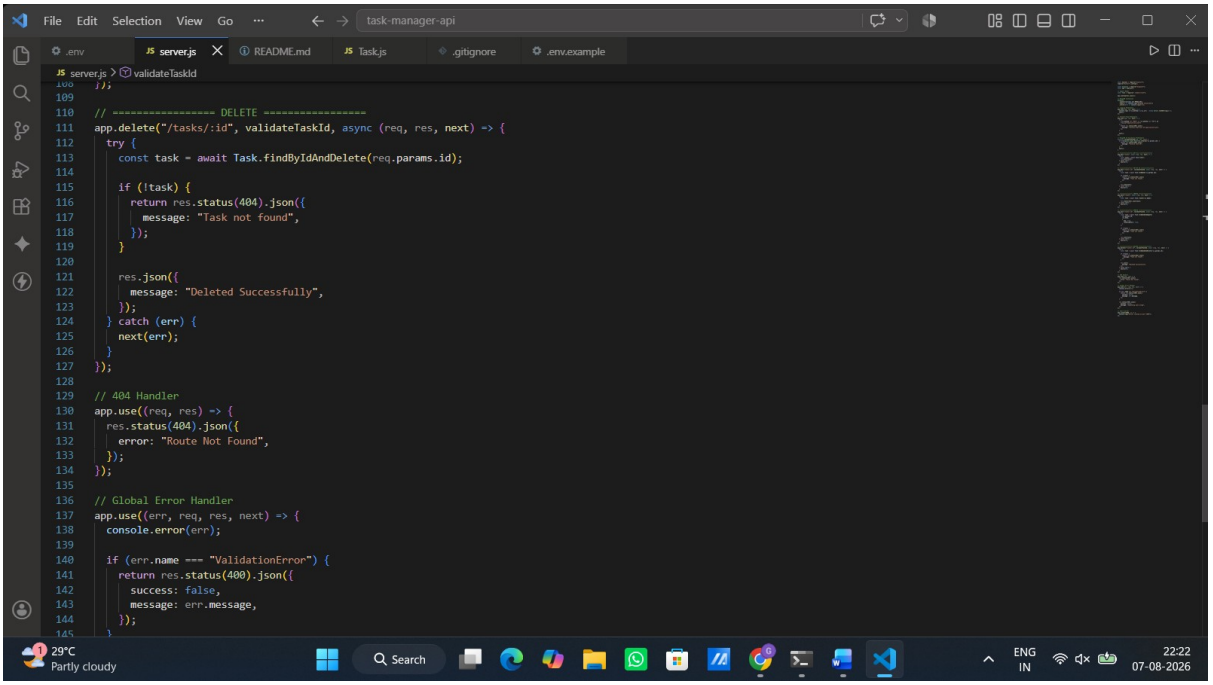
// ================ GET BY ID ================
app.get("/tasks/:id", validateTaskId, async (req, res, next) => {
  try {
    const task = await Task.findById(req.params.id);

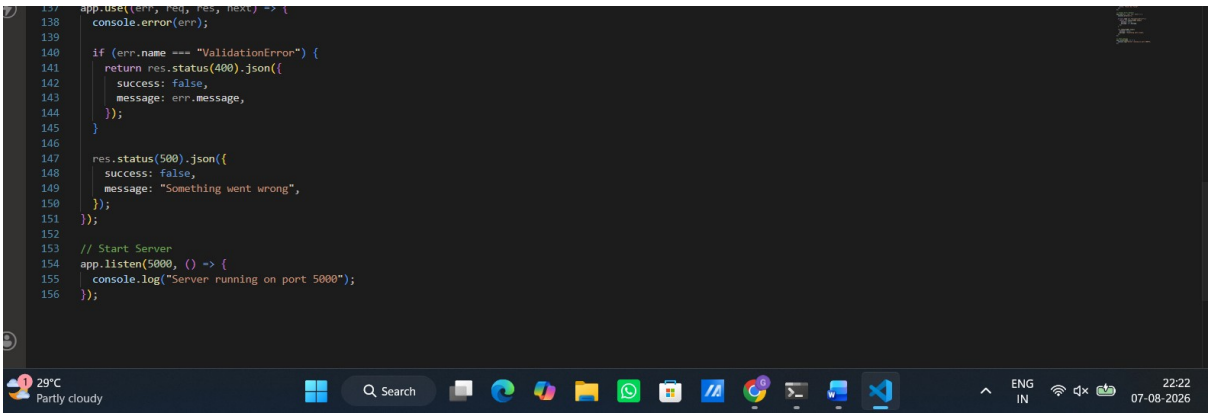
    if (!task) {
      return res.status(404).json({
        message: "Task not found",
      });
    }

    res.json(task);
  } catch (err) {
    next(err);
  }
});

// ================ CREATE ================
```

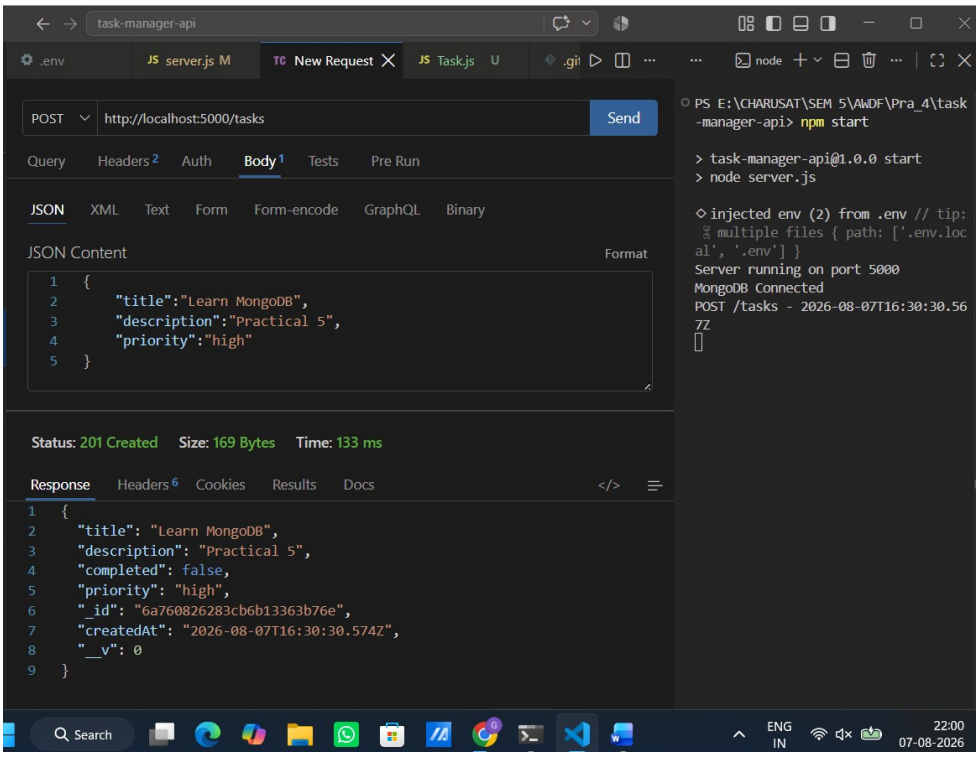

File Edit Selection View Go
task-manager-api
.env
server.js
README.md
Task.js
.gitignore
.env.example
server.js > validateTaskId
77 try {
78 const task = await Task.create(req.body);
79
80 res.status(201).json(task);
81 } catch (err) {
82 next(err);
83 }
84 });
85
86 // ================ UPDATE ================
87 app.put("/tasks/:id", validateTaskId, async (req, res, next) => {
88 try {
89 const task = await Task.findByIdAndUpdate(
90 req.params.id,
91 req.body,
92 {
93 new: true,
94 runValidators: true,
95 }
96);
97
98 if (!task) {
99 return res.status(404).json({
100 message: "Task not found",
101 });
102 }
103
104 res.json(task);
105 } catch (err) {
106 next(err);
107 }
108 });
109
110 // ================ DELETE ================
111 app.delete("/tasks/:id", validateTaskId, async (req, res, next) => {
112 try {
113 const task = await Task.findByIdAndDelete(req.params.id);
29°C
Partly cloudy
Search
ENG
IN
22:21
07-08-2026


File Edit Selection View Go
task-manager-api
.env
server.js
README.md
Task.js
.gitignore
.env.example
server.js > validateTaskId
110 // ================ DELETE ================
111 app.delete("/tasks/:id", validateTaskId, async (req, res, next) => {
112 try {
113 const task = await Task.findByIdAndDelete(req.params.id);
114
115 if (!task) {
116 return res.status(404).json({
117 message: "Task not found",
118 });
119 }
120
121 res.json({
122 message: "Deleted Successfully",
123 });
124 } catch (err) {
125 next(err);
126 }
127 });
128
129 // 404 Handler
130 app.use((req, res) => {
131 res.status(404).json({
132 error: "Route Not Found",
133 });
134 });
135
136 // Global Error Handler
137 app.use((err, req, res, next) => {
138 console.error(err);
139
140 if (err.name === "ValidationError") {
141 return res.status(400).json({
142 success: false,
143 message: err.message,
144 });
29°C
Partly cloudy
Search
ENG
IN
22:22
07-08-2026

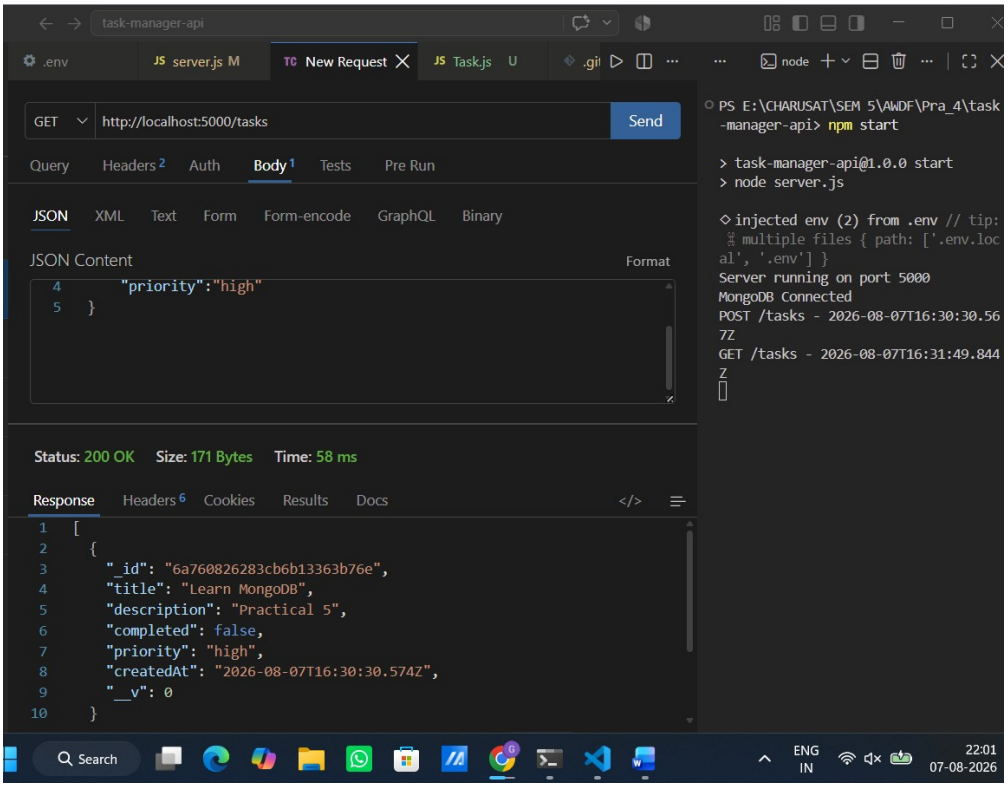


Step 7: Test CRUD API Using Thunder Client

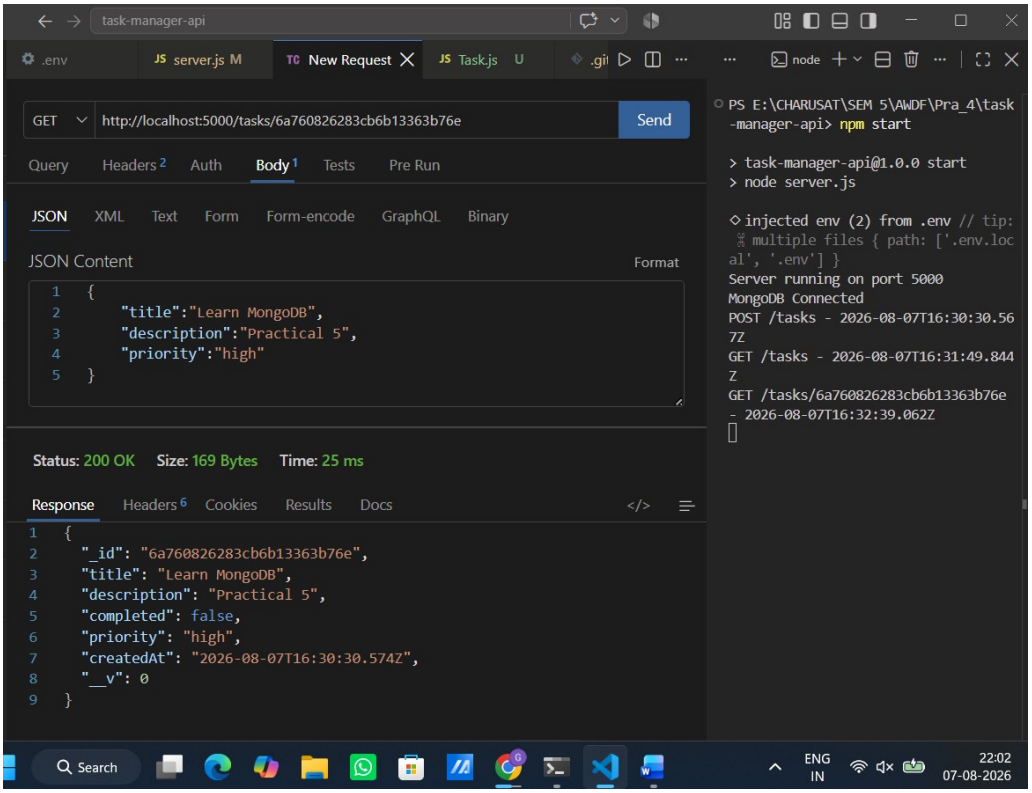
Test 1: Create a Task (POST)



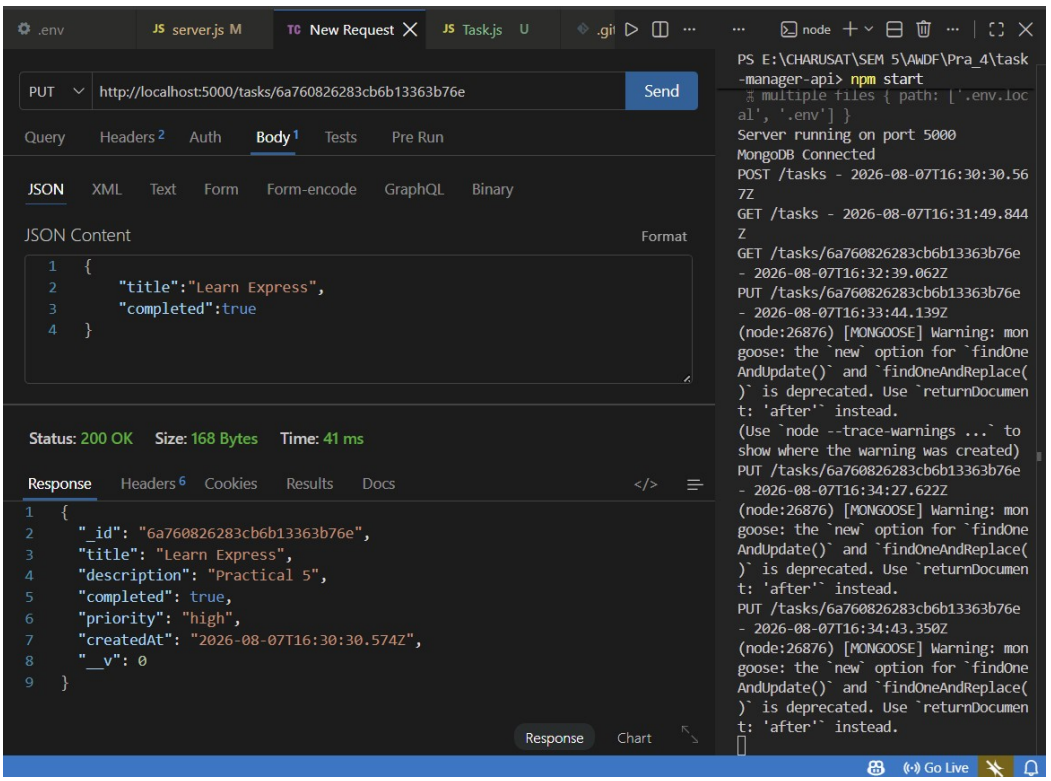
Test 2: Get All Tasks (GET)



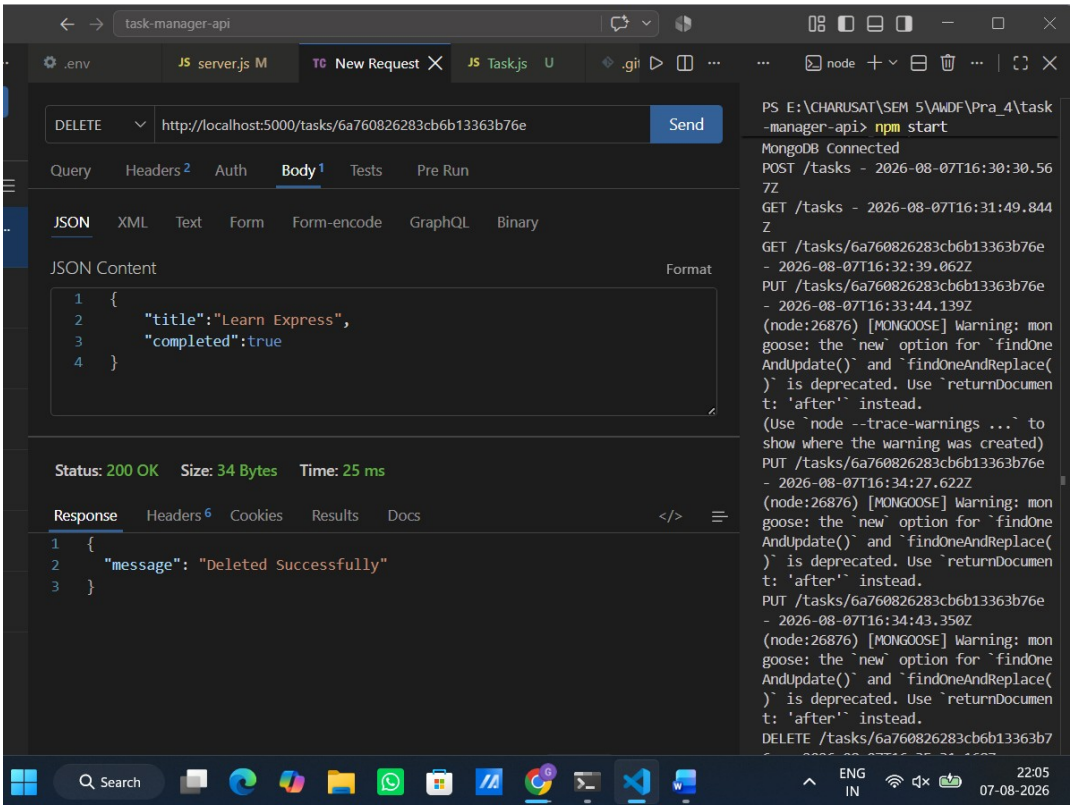
Test 3: Get Task by ID (GET)



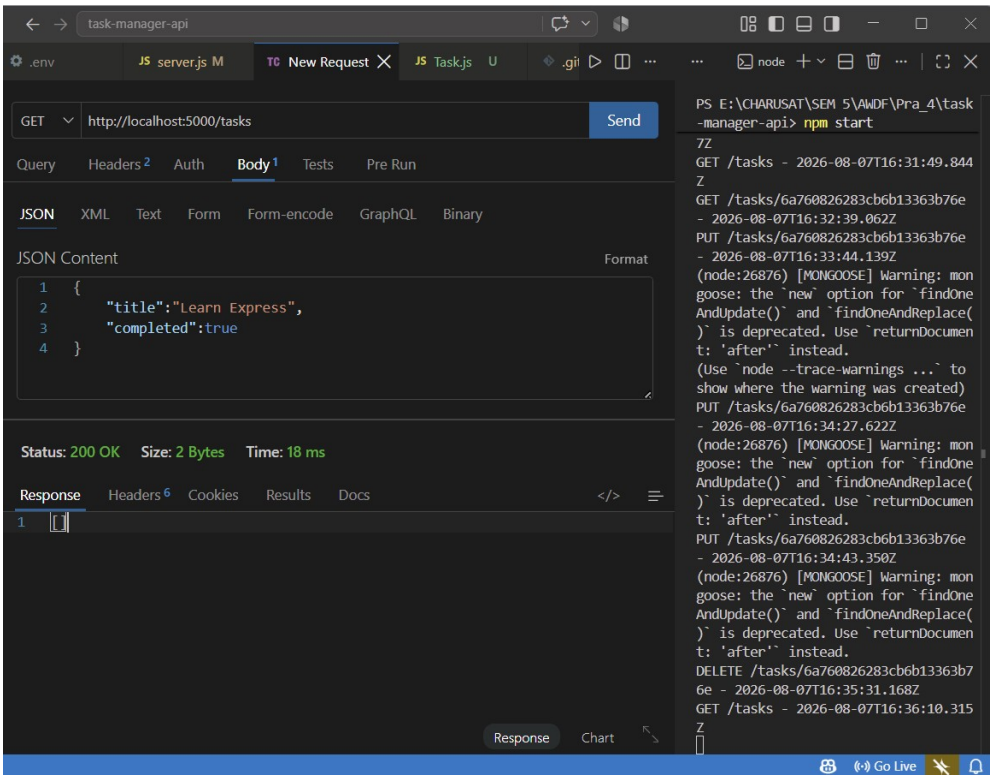
Test 4: Update Task (PUT)



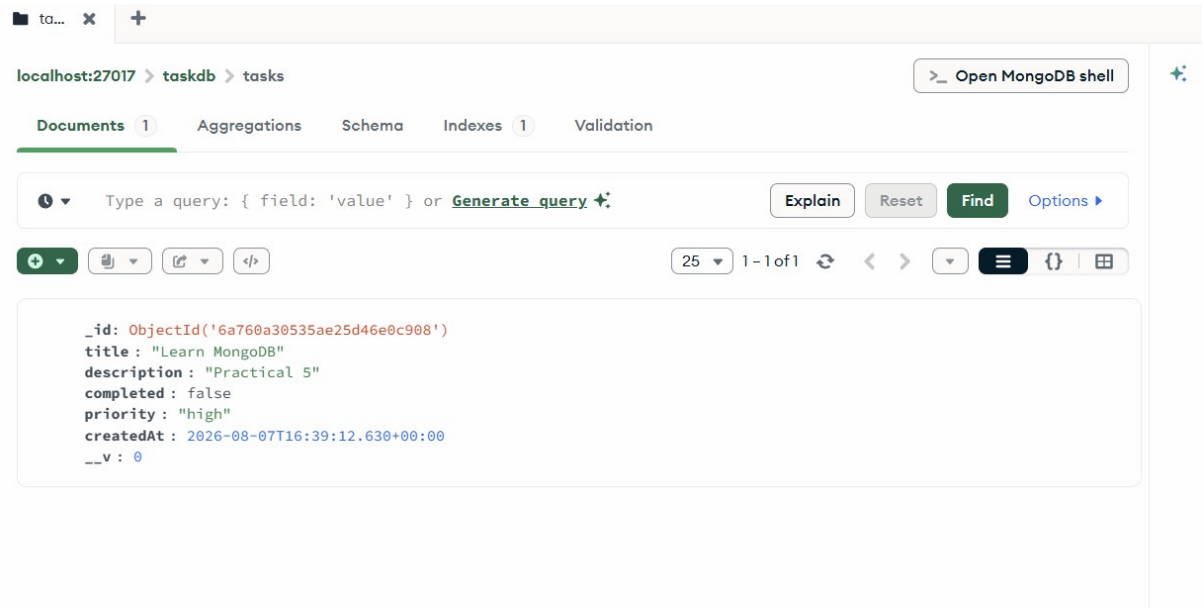
Test 5: Delete Task (DELETE)



Test 6: Verify Deletion



Step 8: Verify Data in MongoDB Compass



Github Repo Link:

https://github.com/sahilmak1602/WDF_24IT045_SAHIL.git